



## Using FlexiConc via CLiC

### A case study on literary fiction

**FlexiConc** is a Python library developed to support concordance analysis for corpus linguists. CLiC incorporates FlexiConc functionality in a dedicated analysis tab. You can find general information on how FlexiConc is used in CLiC [here](#).

In this document, we will use FlexiConc to analyse literary fiction.

This document serves as an overview and illustrates all algorithms that we currently have available. You can either follow the exercises or look at individual algorithms to see examples of how to use them.

Our case study deals with body part nouns in 19<sup>th</sup> century English novels. It aims to find repeated patterns of language use. We use the 19C through the CLiC web app (Mahlberg et al. 2020).

The website can be accessed here: <https://cllc-fiction.com/>

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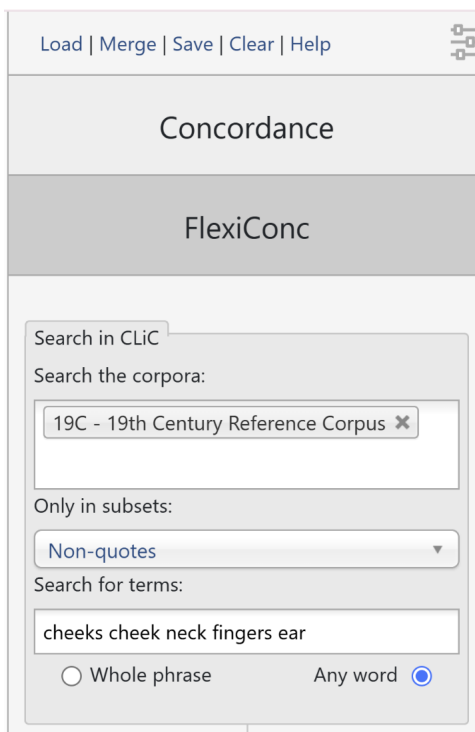
## 1. Running a concordance search in FlexiConc mode

**Background:** The concordance search is the basis for all subsequent steps like sorting or grouping. In the example, we search for a range of body part nouns in *non-quotes* – all parts of the 19C novels outside quotation marks. All further steps in this document are based on the same concordance created here. To use FlexiConc in CLiC, select the FlexiConc tab on the right side of the page.

### Instructions:

1. Set the **corpus** to **19C**, **subset** **non-quotes**, and enter the **search terms** *cheeks cheek neck fingers ear*. The terms should be added with just a space as a separator (no commas or similar – Figure 1).
2. Select the **radio button** *Any word* to search the words separately rather than as a sequence (Figure 1).
3. Click the **confirmation button** to start searching the corpus (Figure 2).
4. You will see the resulting **concordance** in the main window (Figure 3).

Enter your search terms here:



The screenshot shows the FlexiConc interface. At the top, there are links: Load | Merge | Save | Clear | Help. Below this is a tabbed interface with 'Concordance' and 'FlexiConc' tabs. The 'FlexiConc' tab is active. Under the 'Search in CLiC' section, there is a 'Search the corpora:' field with a dropdown menu showing '19C - 19th Century Reference Corpus'. Below this is an 'Only in subsets:' dropdown menu showing 'Non-quotes'. Under the 'Search for terms:' section, there is a text input field containing 'cheeks cheek neck fingers ear'. At the bottom, there are two radio buttons: 'Whole phrase' (unselected) and 'Any word' (selected).

Figure 2: Entering search terms.



Figure 1: Confirmation button.

## This is what your result should look like:

| Showing 1 to 50 of 1,575 entries, |                                                        |         |                                                         |         |        |
|-----------------------------------|--------------------------------------------------------|---------|---------------------------------------------------------|---------|--------|
| 64b4590                           |                                                        |         |                                                         |         |        |
| ID                                | Left                                                   | Node    | Right                                                   | Book    | In bk. |
| 0                                 | inge of crimson flushed the waxen whiteness of her     | cheeks; | not one shadow of brown redeemed the pale insipi        | LadyAud |        |
| 1                                 | past nine o'clock, singing a little Scotch melody, her | cheeks  | tinged with as delicate a pink as the pale hue          | LadyAud |        |
| 2                                 | 's face; the pretty, roseate flush faded out from her  | cheeks, | and left them waxen white, and angry flashes lighte     | LadyAud |        |
| 3                                 | unless, flaxen hue which generally accompanies pink    | cheeks  | and white eyelashes. She had been a rustic beauty,      | LadyAud |        |
| 4                                 | ing about him, kissed the dirty tears from his faded   | cheeks. | "Don't be sorry for me, gran'pa," he said; "I am        | LadyAud |        |
| 5                                 | The unnatural color still burnt like a flame in her    | cheeks; | the unnatural light still glittered in her eyes. The ex | LadyAud |        |
| 6                                 | ter nor half the folks as are growed up." "I Maggie's  | cheeks  | began to flush with triumphant excitement. She tho      | mill    |        |
| 7                                 | lavishing so many warm kisses on it that the waxen     | cheeks  | had a wasted, unhealthy appearance. "Did you eve        | mill    |        |
| 8                                 | red with the roundest of eyes above the reddest of     | cheeks; | but was now entirely defaced by a long career of        | mill    |        |
| 9                                 | much alike as goslings,--a lad with light-brown hair,  | cheeks  | of cream and roses, full lips, indeterminate nose and   | mill    |        |
| 10                                | any, and they ate together and rubbed each other's     | cheeks  | and brows and noses together, while they ate, with      | mill    |        |

Figure 3: Concordance view.

## 2. A first overview: random sort

### Background:

By default, concordance lines are displayed in the following order:

1. Concordance lines are shown in the order in which the search terms were entered.
2. If there is more than one result, the hits are sorted alphabetically based on the name of the book they appear in.
3. If there is more than one result per search term and book, the hits are sorted based on their position within the book.

Random sorting does precisely what the name suggests: the lines are displayed in a sequence that does not follow a systematically motivated order. That said, the order is not entirely random either: The **optional seed** number ensures that every time someone runs the same 'random sort' step on the same data, they will be presented with the same ordering. This helps to make sure that results are **reproducible**.

### Instructions:

1. **Scroll down** in the FlexiConc panel until you see the **add algorithm** button below the tree symbol (Figure 4).
2. **Scroll down** in the selection menu to find the **random sort** function and click on it.
3. The algorithm will run immediately. Keep the **optional seed** that is set to **42** by default (Figure 5).

### Find and use Random Sort like this:

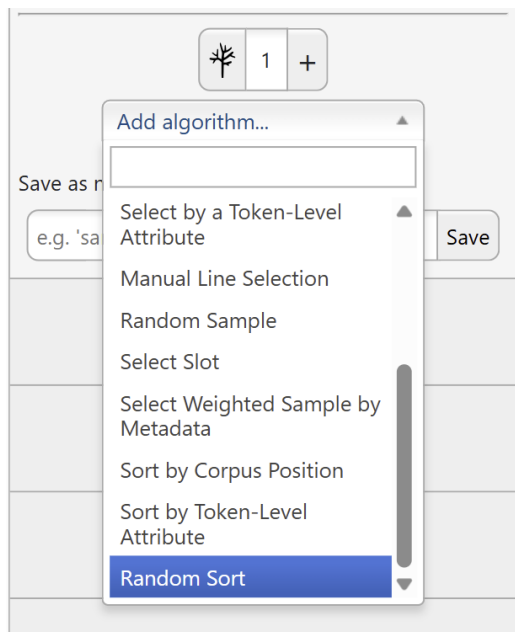


Figure 4: Finding the *Random Sort* algorithm.

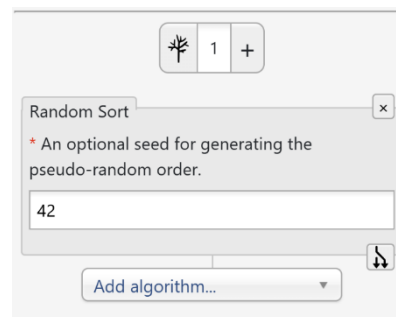


Figure 5: Seed for *Random Sort*

### This is what your result should look like:

|                                                                             |          |                                                                |
|-----------------------------------------------------------------------------|----------|----------------------------------------------------------------|
| change over the counter, in taking which he caught her                      | fingers  | and held them. There was a slight struggle and titter          |
| uttering these three terrible _musts_, Klesmer lifted up three long         | fingers  | in succession. "In sum, you have not been called upon          |
| the music was her only refuge from herself. Sometimes her                   | fingers  | touched the notes with a lingering fondness--a soft, plaintive |
| appalled. His vows of austerity and celibacy echoed in his                  | ear;     | his thirst after holiness--had it been quenched at so          |
| the circle. Madame Carolina first presented her delicate and faintly-rouged | cheek    | to the hump-backed Crown Prince, who scarcely raised his eyes  |
| end over the lady's heart and the other to his                              | ear,     | and listened carefully. ¶ When he had done he says to          |
| her window, and then went to assist Mrs. Thornton's trembling               | fingers. | ¶ From some cause or other, there was a pause of               |
| kind of nightmare; I felt the fiend's grasp in my                           | neck     | and could not free myself from it; groans and cries            |
| which Nature had gifted Lady Crawley were those of pink                     | cheeks   | and a white skin, and as she had no sort                       |
| the high-cornered outline of an early Gothic shield; and a                  | neck     | which was smooth and round as a cylinder. The lower            |

Figure 6: Concordance arranged by *Random Sort*

### 3. Adding information: annotations

**Background:** *Annotation* refers to the automatic analysis of the concordance lines. In CLiC's FlexiConc mode, several types of annotation are available:

- SpaCy POS tags
- Sentence transformers
- SpaCy embeddings
- TF-IDF embeddings

**! Note:** Annotations are different from algorithms: **you will not see immediate changes** to your concordance after running an annotation step. What annotation does is to enable the use of tags or embeddings *in certain algorithms*.

In the following two subsections, we will show you examples of the two most central annotations: POS tagging with spaCy and embeddings with sentence transformers (Reimers & Gurevych 2019).

#### 3.1 POS tagging

**Background:**

**PoS** (part-of-speech) tags are a cornerstone of corpus linguistics. A POS tag, such as *ADJ* for adjectives, represents the morpho-syntactic category of a token. This information can be used in place of tokens in all algorithms that operate on *token-level attributes*.

In the FlexiConc integration of the CLiC web app, the concordance can be annotated with POS tags from the NLP library spaCy (<https://spacy.io/usage/linguistic-features>). POS tagging is supported as an annotation layer that is added to the concordance. After doing so, we can use the tags in all algorithms that rely on token-level attributes.

In the following example, we add POS tags using the **en\_core\_web\_md** model. SpaCy also offers different models and additional grammatical information such as dependency. Also, note that the labels used for POS tagging may differ depending on the models used. For details, please refer to the spaCy documentation.

**Instructions:**

1. Find the **add annotation** menu.
2. Select **annotate with spaCy POS tags** (figure 7).  
 ! FlexiConc supports various types of annotations that produce different results. Make sure to choose *SpaCy POS tags* and not, e.g., *SpaCy embeddings*.
3. Select the **model** *en\_core\_web\_md* (figure 8)  
 ! CLiC currently offers two English and one German tagging model. They are sorted alphabetically, with the German one (*de\_core\_news\_sm*) active by default. When working on English data, make sure to change this selection to one of the *en\_* models.
4. To add POS tagging, **select the attribute** *pos\_* from the dropdown menu of spaCy token attributes (figure 9).
5. Click **confirm** in the concordance window.

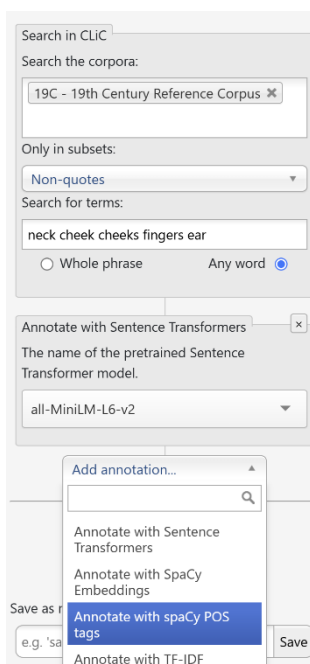
**Find and annotate spaCy POS tags like this:**

Figure 7: *Annotate with spaCy POS tags.*

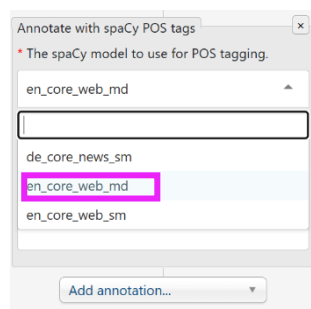


Figure 8: *spaCy models:* *de\_* is for German, *en\_* for English.

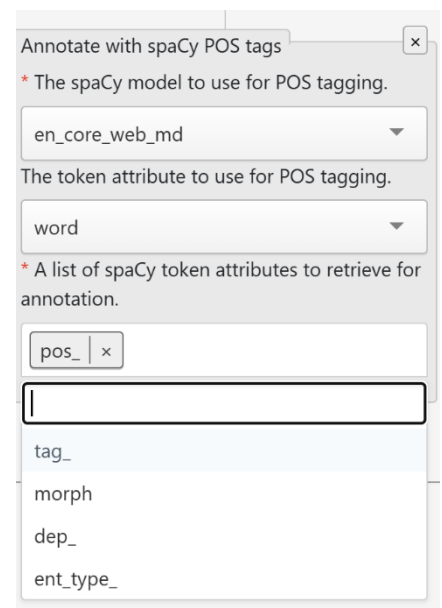


Figure 9: Token attributes for annotation

We are not showing you a screenshot of the concordance output here, because it will not change from adding an annotation. Instead, the result of this annotation is that you can now use the *pos\_* attribute in algorithms that require token-level attributes. For example, *Select by Token-Level String Attribute* will now allow you to select lines containing adjectives, as shown in [Step 4.2](#).

## 3.2 Embeddings

### Background:

**Embedding** annotations form the basis for *clustering* algorithms (cf. section 5). Clustering is used in FlexiConc to form partitions based on the similarity between concordance lines. To perform clustering, the desired annotation must be added beforehand.

As illustrated in the second screenshot below, embedding algorithms can use a **context window** with upper and lower bounds. For instance, you can set the upper bound to -1 to calculate similarities based on the left context.

Each type of embedding annotation represents a different method for determining similarity scores. While TF-IDF and spaCy embeddings are relatively established, sentence transformers (Reimers & Gurevych 2019) are a more recent method that draws on data from a pre-trained transformer model.

In this example, we add annotations based on sentence transformers.

**! Note:** As with the annotation for POS tags in the previous step, this annotation will not produce a visible difference to your concordance view. Instead, it is the prerequisite for the algorithm [Flat Clustering by Embeddings](#).

### Instructions:

1. Click on the **add annotation** button below the query menu (Figure 10).
2. Select **annotate with Sentence Transformers** (Figure 10).
3. Click **confirm** in the concordance window.
4. Available options are shown in Figure 11- for our running example, leave them as-is.

## Find and annotate sentence transformer embeddings like this:

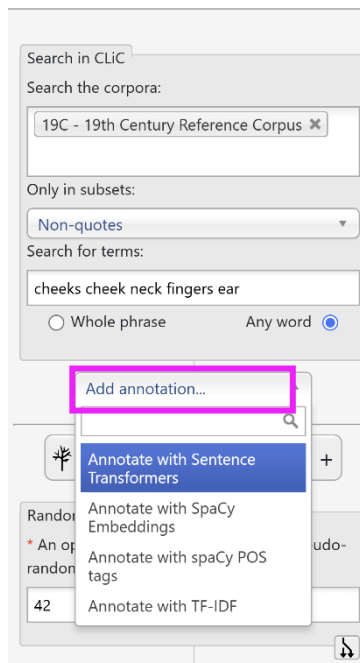


Figure 10: *Annotate with Sentence Transformers*

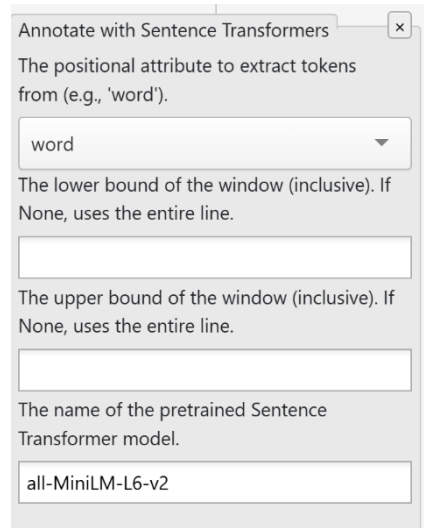


Figure 11: Parameter settings for *Annotate with Sentence Transformers*

As mentioned in the note above, the concordance output will not change from adding an annotation. Instead, the result of embedding annotations is that you can now use the resulting similarities for clustering with the algorithm *Flat Clustering by Embeddings*. This will allow you to partition your concordance into lines with high similarity scores, as shown in [Step 5](#).

## 4. Zooming in: Select by Token-Level String Attribute

### Background:

*Select by token-level string attribute* creates a **subset** of the concordance. This means that the result will only include lines fulfilling the criteria specified by the algorithm settings.

The criteria imposed by this algorithm are based on the characteristics of the words in a specific position relative to the node. Most often, this algorithm will be used to specify a particular word immediately before or after the node term.

To specify the position, we use the **offset** parameter, where 0 corresponds to the node. Negative offsets pertain to the left context, positive offsets to the right context.

The *positional attribute* is typically set to **word**, which represents the literal word-form, or token, itself (see [4.1](#) for an example). If you have token-level annotations like POS tags (see [3.1](#) for adding POS annotations), you can also choose them as a selection criterion (see [4.2](#)).

**! Note:** The **checkboxes** below the offset menu offer additional settings that may be less common:

- **Case-sensitive** matching: *her* and *Her* are counted as different tokens.
- **Regex** matching: the value is interpreted as a regular expression. Using this option, you could specify that the token on offset position -1 has to end in *-ing* by selecting *+ing\b*.
- **Inverting** the selection: select all lines where the token on offset position -1 is **not** *her*.

## 4.1 Selecting based on the word form

**Background:** *Select by token-level string attribute* based on the **word** is by far the most common use-case of this algorithm: lines are selected if the word in a certain offset position corresponds to the values we entered in the *string values* window.

In the following example, we select concordance lines where the token immediately to the left of the node (= the body part noun) is *her*. The word *her* ('string attribute') is specified as a string value, and the position -1 corresponds to the 'token offset'.

As a result, the subset will contain all uses of *her cheeks*, *her cheek*, *her neck*, *her fingers*, and *her ear*.

### Instructions:

1. Find *Select by token-level string attribute* in the *add algorithm* menu (Figure 12)
2. Set the **value** to match against to **her**
3. The **token attribute** is set to **word** by default – keep it that way.
4. Set the **offset** to **-1**

Find and use *Select by Token-Level String Attribute* like this:

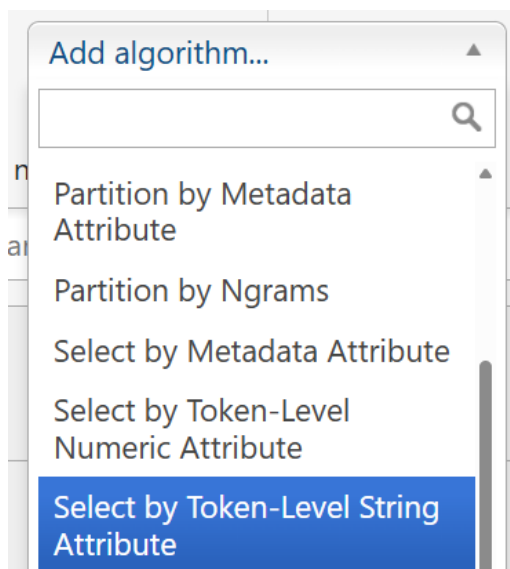


Figure 12: *Select by Token-Level String Attribute*

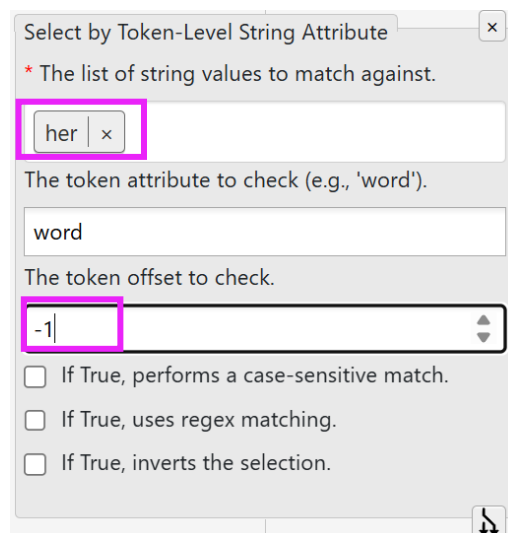


Figure 13: Parameter settings for *Select by Token-Level String Attribute* - word

### This is what your result should look like:

|                                                                               |                                                                |
|-------------------------------------------------------------------------------|----------------------------------------------------------------|
| change over the counter, in taking which he caught <b>her</b> fingers         | and held them. There was a slight struggle and titter          |
| the music was her only refuge from herself. Sometimes <b>her</b> fingers      | touched the notes with a lingering fondness--a soft, plaintive |
| a round full face and a high colour in <b>her</b> cheeks.                     | The second sister was Fanny, a very pretty little girl         |
| took out her handkerchief, and wiped the spot on <b>her</b> cheek             | that had been touched by his lips. His ardour was              |
| she began to sob, and the tears rolled down <b>her</b> cheeks--               | a form of agitation which she had never shown before           |
| flush with which warmth and sound sleep had suffused <b>her</b> cheeks        | and neck dissolved from view, and the deathlike pallor in      |
| Hale imagined that he saw traces of tears on <b>her</b> cheeks.               | But that could not be, for she brought good news               |
| indeed, thy fate!' whispered again the Egyptian's voice in <b>her</b> ear.    | 'And thou art destined to be the bride of Arbaces              |
| ask me why her breath is sweet and why <b>her</b> cheek                       | is blooming, It is because the sun is out and                  |
| played caressingly with her silken hair--his breath fanned <b>her</b> cheek-- | lone, the cruel lone, was not by--none other demanded          |

Figure 14: *Select by Token-Level String Attribute*: concordance of *her* on token offset -1

## 4.2 Selecting based on the POS tag

**Background:** Selecting by token-level string attributes other than the word form (token) itself is very similar to the step described above: Instead of specifying that the word in a certain position must correspond to a form like *her*, we can restrict the same position to a category such as *adjective*.

To do this, you simply need to change the positional attribute to the POS annotation (see section [3.1](#) for how to annotate your concordance with POS tags).

We will now use the spaCy tags added in [step 3.1](#) to select concordance lines based on grammatical information. In the example, we create a subset where the word immediately left to the node (offset -1) is an adjective (ADJ). As a result, we will get lines with *cold fingers*, *rosy cheeks* etc.

In the following example, we select concordance lines where the token immediately to the left of the node (= the body part noun) is *her*. The word *her* ('string attribute') is specified as a string value, and the position -1 corresponds to the 'token offset'.

As a result, the subset will contain all uses of *her cheeks*, *her cheek*, *her neck*, *her fingers*, and *her ear*.

**! Note:** If you are following the worked example in this document, you will currently be working on a subset of the concordance limited to *her* on position -1. Applying this additional subset to the same lines will result in an empty subset, as there cannot be lines that feature both *her* and ADJ in the same position. Therefore, you'll need to **create a new branch** (see [section 6](#)) before applying this second selection algorithm!

This new branch will be based on the full initial concordance, so that we can create a new subset.

### Instructions:

1. Select a new algorithm **Select by Token-Level String Attribute** in the *add algorithm* menu (Figure 12).
2. Set the **value** to match against to **ADJ**
3. Set the **token attribute** to **pos\_**
4. Set the **offset** to **-1**

**Find and use Select by Token-Level String Attribute like this:**

Figure 15: Parameter settings for *Select by Token-Level String Attribute* – POS tag

## This is what your result should look like:

|                                                                                   |                                                                    |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------|
| were of that sunless, flaxen hue which generally accompanies <b>pink</b> cheeks   | and white eyelashes. She had been a rustic beauty, perhaps         |
| for company, and they ate together and rubbed each <b>other's</b> cheeks          | and brows and noses together, while they ate, with a               |
| with a pair of amiable blue-gray eyes and round <b>pink</b> cheeks                | that refused to look formidable, let him frown as he               |
| she was, the girl's Roman blood rushed over her <b>pale</b> cheeks                | as she felt the insult. She turned towards Hermanric, looked       |
| She started suddenly, a bright flush flew over her <b>colourless</b> cheeks;      | she bent down, and looked earnestly and wistfully into the         |
| the Prefect, staring round him with bewildered eyes and <b>colourless</b> cheeks. | 'Our miseries are but dawning as yet! In the next                  |
| of the stranger was of a ghastly paleness; his <b>hollow</b> cheeks               | were seamed with deep wrinkles; and his eyes glared with           |
| and a deadly paleness overspread the once plump and <b>jovial</b> cheeks          | which so many mistresses had kissed in mercenary rapture in        |
| tresses drooped heavily on either side of the white <b>hollow</b> cheeks.         | 'Behold!' cried Vetrano, pointing to the corpse--'Behold my secret |
| days. The cold fingers that trembled round mine--the <b>pale</b> cheeks           | with a bright red spot burning in the midst of                     |

Figure 16: *Select by Token-Level String Attribute*: concordance of adjectives on token offset -1

## 5. Investigating similarity: Flat Clustering by Embeddings

**Background:** Clusters are groups of concordance lines that are more similar to one another than to the remaining lines in the concordance. In the clustering methods FlexiConc supports in CLiC, the analyst determines how many groups are formed. The clustering algorithm produces precisely that number of concordance groups so that similar lines go together.

To perform clustering, you first need to annotate the concordance with one of the available embedding methods (cf. [section 3.2](#)). The similarity scores resulting from this annotation serve as the basis for the clustering algorithm to form groups.

### ! Notes:

1. If you are following the running example, add this algorithm to **branch 1** – that is, the lines selected for *her* on -1.
2. In the example, we use k-means clustering. FlexiConc also offers agglomerative clustering. In our experiments so far, k-means appears to work better, as the resulting clusters are more similar in size. That said, clustering for concordance lines is not yet a common approach and agglomerative clustering is worth exploring. For the agglomerative method, you can set the *distance metric* and *linkage* criterion. Under the hood, FlexiConc uses the clustering algorithms implemented in scikit-learn.

On the scikit-learn website, you can find more details about how the different clustering algorithms work: <https://scikit-learn.org/stable/modules/clustering.html>.

### Instructions:

1. Select a new algorithm **Flat clustering by embeddings** in the *add algorithm* menu (Figure 17).
2. Set the **metadata column** to **embeddings\_sentence\_transformers** (Figure 18). If you cannot find this option, add the annotation as shown in [step 3.2](#).
3. Change the **number of clusters** to **10**. This setting determines how many groups are formed – changing it can strongly influence the results (Figure 18).

As a result of this step, all occurrences of *her* + node will be partitioned into 10 clusters of similar concordance lines.

### Find and use Flat Clustering by Embeddings like this:

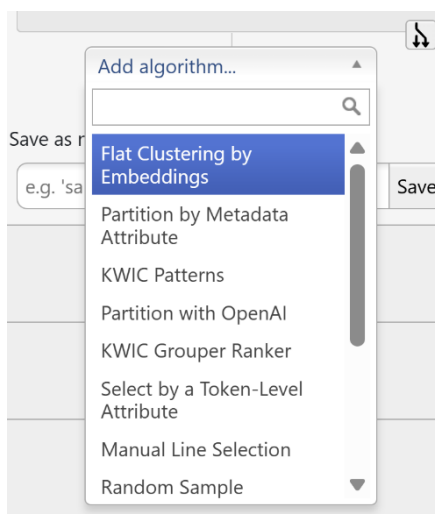


Figure 17: Flat Clustering by Embeddings

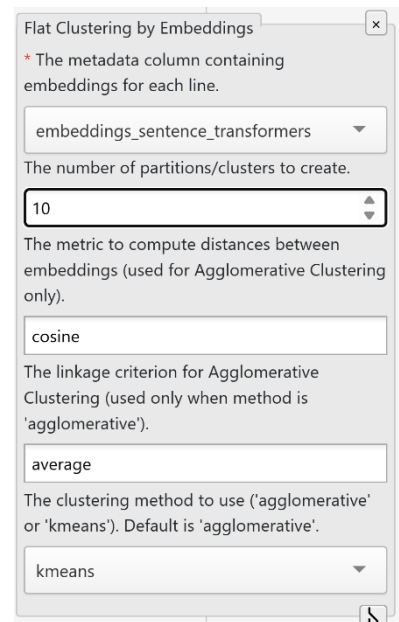


Figure 18: Flat Clustering by Embeddings: Parameter settings

### This is what your result should look like:

On running the steps above, your concordance will be *partitioned* into clusters. By default, the first cluster (called cluster\_0) is shown, with the remaining clusters collapsed (see Figure 19).

| Left                                                                 | Node                                              | Right                                                                | Book    | In bk.   |
|----------------------------------------------------------------------|---------------------------------------------------|----------------------------------------------------------------------|---------|----------|
|                                                                      | Partition cheeks cheek neck fingers ear Cluster_0 |                                                                      |         | 56 lines |
| quiet and still for some minutes. But, on stroking her               | cheek                                             | in a caressing way soon after, he was almost shocked                 | NorthS  |          |
| detect her raising a hand, and brushing something off her            | cheek.                                            | I gazed round for a means of diverting her thoughts                  | wh      |          |
| by her long hair and drove the knife into her                        | neck.                                             | The moaning accents of the girl, bewailing her approaching fate      | Antoni  |          |
| a more cheerful way, lifting her hands backward to her               | neck                                              | and moving toward the door. She wanted to waive those                | Deronda |          |
| held by the hand, was now being flung over her                       | neck,--                                           | all this yeasty mingling of dimly understood facts with vague        | Deronda |          |
| foreseen that it would end with throttling fingers on her            | neck.                                             | After a pause, she said in the tone rather of                        | Deronda |          |
| girl, as she spoke, touched a talisman suspended round her           | neck.                                             | "Thine own danger!" what danger threatens me? "I 'Had the earthquake | Pomp    |          |
| half-angry gesture, and twisting it backward and forward between her | fingers.                                          | "I think some people are born to be unlucky, Mrs                     | LadyAud |          |
| become upon the now familiar way! "My lady twisted her               | fingers                                           | in her loose amber curls, and made as if she                         | LadyAud |          |
| watch in her hand with the many-doubled chain round her              | fingers,                                          | and observed to Mrs. Tulliver, who had just returned from            | mill    |          |

Figure 19: Clustering view: Cluster 0 uncollapsed

Click on the partition label (marked in Figure 19) to go to an overview of all clusters (Figure 20).

In the overview of clusters (Figure 20), the number to the right shows the number of lines contained in each cluster. This shows us that clusters 0 and 1 are significantly larger than cluster 3.

|           | Node                          | Right     | Book | In bk.   |
|-----------|-------------------------------|-----------|------|----------|
| Partition | cheeks cheek neck fingers ear | Cluster_0 |      | 56 lines |
| Partition | cheeks cheek neck fingers ear | Cluster_1 |      | 50 lines |
| Partition | cheeks cheek neck fingers ear | Cluster_2 |      | 37 lines |
| Partition | cheeks cheek neck fingers ear | Cluster_3 |      | 22 lines |
| Partition | cheeks cheek neck fingers ear | Cluster_4 |      | 39 lines |
| Partition | cheeks cheek neck fingers ear | Cluster_5 |      | 22 lines |
| Partition | cheeks cheek neck fingers ear | Cluster_6 |      | 44 lines |
| Partition | cheeks cheek neck fingers ear | Cluster_7 |      | 49 lines |
| Partition | cheeks cheek neck fingers ear | Cluster_8 |      | 49 lines |
| Partition | cheeks cheek neck fingers ear | Cluster_9 |      | 34 lines |

Figure 20: Overview of cluster partitions

Clustering is a fully automated method, and the results may vary in how useful they are. On the other hand, clusters can help to identify complex patterns in the data.

Clicking on clusters 2 (Figure 21) and 3 (Figure 22) shows that both contain mostly lines for *cheeks* – but with a markedly different usage pattern:

| Partition                                               | cheeks cheek neck fingers ear | Cluster_2                                                               |         | 37 lines    |
|---------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------|---------|-------------|
| chair, with the big tears ready to roll down her        | cheeks,                       | quite blinded to the presence of Bob, who was looking                   | mill    | <div></div> |
| feet were uncovered; the tears still streamed over her  | cheeks.                       | She advanced with the greatest caution (as if fearful of                | Antoni  | <div></div> |
| m her eyes. The tears streamed incessantly down her     | cheeks,                       | but she never turned towards Goisvintha, never suspended her occupation | Antoni  | <div></div> |
| use. No words escaped her, no tears flowed over her     | cheeks                        | when she reascended the steps--hope, fear, thought, sensation itself    | Antoni  | <div></div> |
| of MY hopes too?" ¶ The tears flowed fast down her      | cheeks.                       | She rested one trembling hand on the table to steady                    | wwhite  | <div></div> |
| ew tears gathered in her eyes, and dropped over her     | cheeks                        | slowly as she paused and waited for his answer. He                      | wwhite  | <div></div> |
| the silence, I think; for the tears rolled down her     | cheeks                        | like rain, at these words, and she could not answer                     | cran    | <div></div> |
| wiping a few tears away that had come on her            | cheeks                        | she knew not how, she went out to open the                              | NorthS  | <div></div> |
| Hale imagined that he saw traces of tears on her        | cheeks.                       | But that could not be, for she brought good news                        | NorthS  | <div></div> |
| tter tears of mortification rising and rolling down her | cheeks.                       | No one had ever before dared to treat her with                          | Deronda | <div></div> |
| without speaking, and the tears were rolling down her   | cheeks                        | as she looked at her friends. ¶ "Now, now, Mab!" said                   | Deronda | <div></div> |
| she began to sob, and the tears rolled down her         | cheeks--                      | a form of agitation which she had never shown before                    | Deronda | <div></div> |

Figure 21: Cluster 2 from the worked example

| Partition                                             | cheeks cheek neck fingers ear | Cluster_3                                                          | 22 lines |
|-------------------------------------------------------|-------------------------------|--------------------------------------------------------------------|----------|
| adroit little suck to the interior of each of her     | cheeks                        | in succession, by which curious and original manoeuvre she brought | Jude     |
| with conflict. It did not bring the blood to her      | cheeks,                       | but it sent it away from her lips. She controlled                  | Deronda  |
| ay." ¶ Gwendolen's lips were almost as pale as her    | cheeks;                       | her passion had no weapons--words were no better than              | Deronda  |
| uch like young animals, and so she could rub her      | cheek                         | against his, and kiss his ear in a random sobbing                  | mill     |
| rother's; baby's little fat hand, pressed against her | cheek,                        | seemed to show more strikingly that the cheek was faded            | mill     |
| in a close embrace, to imprint one kiss on her        | cheek                         | or forehead? I was not unreasonable--that was all I                | Prof     |
| gure, her long white throat, and the curves of her    | cheek                         | and chin were always set off to perfection by the                  | Deronda  |
| g down by her with circling arms, she pressed her     | cheek                         | against Gwendolen's head, and then tried to draw it upward         | Deronda  |
| ad vanity. One day, indeed, he had kissed not her     | cheek                         | but her neck a little below her ear; and Gwendolen                 | Deronda  |
| many changes besides those in the contour of her      | cheek                         | and throat; and that Grandcourt should marry her had become        | Deronda  |
| Sybil rested on the arm of her chair, and her         | cheek                         | upon her hand; as Egremont said these words she shaded             | sybil    |
| er agitated countenance, a tear coursed down her      | cheek                         | big as the miserable Medora's! ¶ CHAPTER VIII. ¶ On the evening    | VivianG  |
| ib and kissed her: her forehead was cold, and her     | cheek                         | both cold and thin, and so were her hand and                       | jane     |

Figure 22: Cluster 3 from the worked example

## 6. Branching out: moving in the analysis tree

### Background:

The analysis tree is a key feature of FlexiConc. It documents all the algorithms we apply. The tree provides two key advantages over traditional concordance tools:

- **Research documentation:** the tree tracks all steps of your analysis, and the tree itself can be shared and restored (see [section 7](#)).
- **Supporting the analysis procedure:** the tree lets you return and build on previous steps in the analysis: see the example in this section.

The tree can be very useful during the analysis. We can quickly return to a previous step and take alternative routes by creating new branches. For instance, sections 4.1 and 4.2 contain two selection algorithms based on different criteria that were created by branching off from the same node in the tree.

In the present example, we create a branch to examine *ADJ + body-part noun* in the same way that we selected concordance lines for *her + body-part noun*. This section shows you how to create a branch from an existing node. The selection step itself is shown in [4.2](#).

**! Note:** The FlexiConc integration in CLiC offers two distinct ways to create branches. Both produce equivalent results, but they are suited for slightly different purposes:

- 1) The **arrow branch symbol** (Figure 23) produces a new branch that includes a copy of all steps in your old branch up to the step from which you 'branched off'. Any algorithms performed in the active branch below the node from which you branch off (in our case, *flat clustering by embeddings* from [section 5](#)) are not copied over from the 'old' branch. This branching technique lets you carry over selected steps and lets you explore 'alternative angles'.
- 2) The **plus symbol** (Figure 24) creates a fully empty top-level branch without any algorithms. This lets you "start from scratch" in a new branch while keeping the old branches intact.

*Regardless of how you create a new branch, you can always switch to your previous branch by clicking 1 next to the tree symbol (Figure 24).*

### Instructions:

1. **Scroll up** to the node *Random sort* with the seed 42.
2. **Click** on the **branch** icon – the two crossing arrows in the bottom right corner below the seed field.
3. Follow the instructions in [4.2](#) to add a new selection based on POS tags using *Select by a token-level string attribute*.

Create and view a new branch like this:

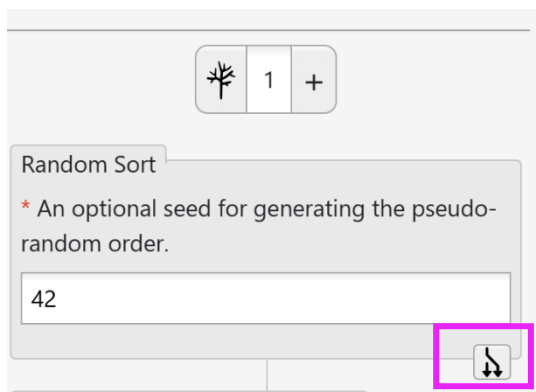


Figure 23: Branch symbol

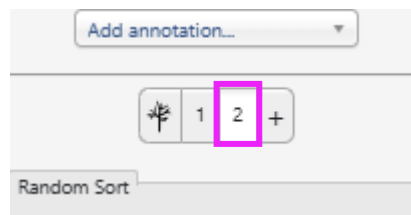


Figure 24: Your newly created branch shows up as "2"

This is what your result should look like:

This step creates a **new branch** where all steps before the one you branched off from are the same.

**! Note:** When using FlexiConc **within the same concordance** analysis, all your steps are automatically stored in the analysis tree.

However, if you

- change your concordance to a different query, or
- leave the website,

your steps will not be saved by default.

Therefore, **you should always save your work** when you are done ([step 16](#)).

## 7. The big picture: tree view

### Background:

If you have been following the entire example, you now have **two branches** in your analysis.

The first branch was created when you applied the very first algorithm, Random Sort, in [section 2](#). The second branch was created in [section 6](#), when you selected by POS tag as shown in [section 4.2](#).

As the analysis becomes increasingly complex, it is useful to look at the **tree structure** as a whole. The tree view mode visualises all steps across all branches.

### Instructions:

1. Click on the **tree symbol** (Figure 25) to see a visualization of the entire analysis (Figure 26).
2. In the tree overview (Figure 26), **click on 2** to switch to branch 2 (containing lines with *adjectives* to the left of the node).

Access the tree view like this:

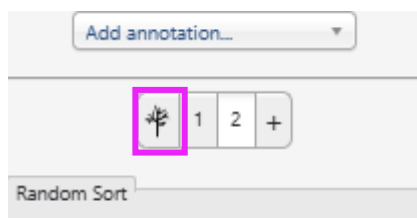


Figure 25: Tree Symbol

This is what your result should look like:

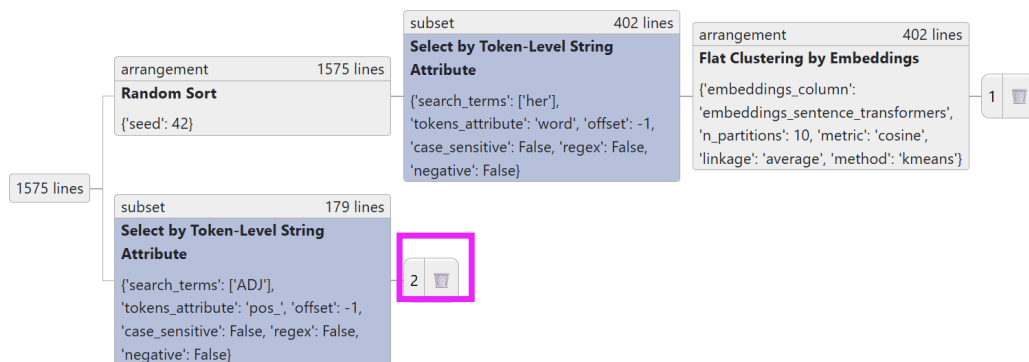


Figure 26: Analysis Tree

### ! Note:

The tree (Figure 26) shows all steps taken so far:

- The first box, **1575 lines**, represents the initial query.
- Branch 1 consists of three nodes:
  - The second box, **Random Sort**, represents the first algorithm.
  - Random Sort is followed by **Select by Token Level String Attribute** to select *her* on position -1.
  - The selected lines were then partitioned via **Flat Clustering by Embeddings**.
- Branch 2 contains a single node, **Select by Token Level String Attribute** with pos\_ set to ADJ on -1. That is, we applied the same algorithm type in the same position twice, but on different branches.

By clicking on the numbered boxes at the end of each branch, you can switch back to that stage of the analysis.

## 8. Rearranging Alphabetically: Sort by Token-Level Attribute

### Background:

Alphabetical sorting is one of the main methods to identify concordance patterns.

In FlexiConc, the corresponding algorithm is *Sort by Token-Level Attribute*. Like *Select by Token-Level Attribute* ([section 4](#)), sorting can be applied to the token itself (default) or to any annotation on the token layer (like [POS tags, section 3.1](#)) after annotating the concordance.

In branch 2 of our example, we selected lines based on the POS tag on the position -1. As a result, the token left to the node can be any word that was tagged as an adjective.

The words that precede the node in this subset share a grammatical category, but the word form itself will differ between lines.

To see whether these adjectives repeat, we can use alphabetical sorting on the word level.

### Instructions:

1. Select the algorithm *Sort by Token-Level Attribute* (Figure 27).
2. Set the *offset* to -1 – the position one token to the left of the node (Figure 28).

### ! Note:

By changing the *context* setting in the second drop-down menu, you could also sort on the entire left or right context (= everything before/ after the node).

For our present case, however, sorting on a specific token is more useful to see which adjectives are repeated (e.g. *ashy cheeks*, see Figure 27).

Find and use Sort by Token-Level Attribute like this:

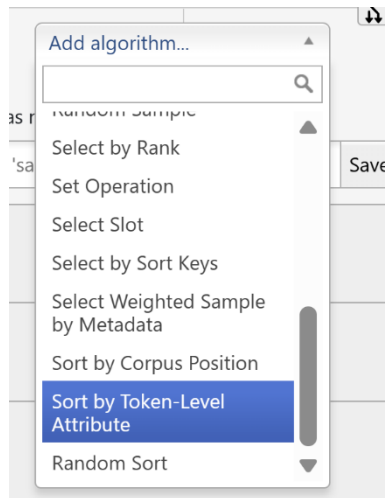


Figure 27: Sort by Token-Level Attribute

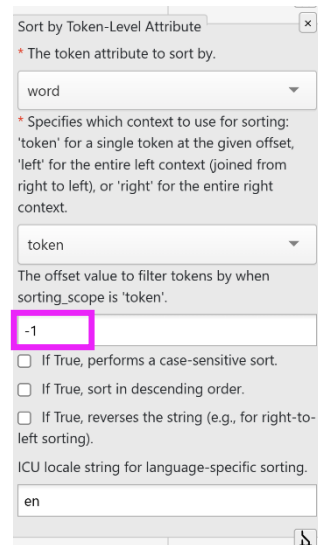


Figure 28: Parameter settings for Sort by Token-Level Attribute

This is what your result should look like:

Showing 1 to 50 of 179 entries,

54b4590

| ID   | Left                                                                       | Node     | Right                                                              |
|------|----------------------------------------------------------------------------|----------|--------------------------------------------------------------------|
| 1451 | herself crouching in her hitherto pure waters, while her <b>anxious</b>    | ear      | listens for the bounding step of the regardless huntsman. ¶ "The   |
| 102  | from his mother rose dark and slow in his <b>ashy</b>                      | cheeks.  | ¶ "Is that woman," he asked, "the woman whom you once              |
| 189  | the narcotic. Again some return of colour to the <b>ashy</b>               | cheeks,  | and the regular breathing of healthy sleep. This time I            |
| 1274 | of the rapid Tiber now caught his suspicious and <b>attentive</b>          | ear.     | They still moved onward a few yards; and then paused               |
| 1446 | bending over her with starting eyes, once more the <b>attentive</b>        | ear      | listening for the soundless breath. No sound! not even a           |
| 611  | up and struck furiously with the knife at her <b>bare</b>                  | neck.    | As the weapon descended, Hermanric caught her wrist. She struggled |
| 99   | look, aided by his light hair and his plump <b>beardless</b>               | cheeks,  | his easy manner and his ever-ready smile, his eyes which           |
| 425  | were a tender inflexibility, the tears trickling down her <b>beautiful</b> | cheek,   | and pressing his hand in both of hers, she subdued                 |
| 1184 | have!" Mr. Osborne said to his clerk, snapping his <b>big</b>              | fingers, | and jingling all the guineas and shillings in his great            |
| 112  | observed that a warm flush now rosed her hitherto <b>blue</b>              | cheeks,  | and felt that her hanging hand was no longer cold                  |
| 519  | with her golden ringlets flying loose behind, and her <b>bright</b>        | cheek,   | as soft and pure in its bloom as a wild                            |
| 301  | left no obvious trace; the eyes are liquid, the <b>brown</b>               | cheek    | is firm and round, the full lips are red. With                     |
| 973  | responsively at every casual movement of the usher's supple <b>brown</b>   | fingers, | and every passing distortion of the usher's haggard yellow face    |
| 1236 | the bridge. And that queer white cur with the <b>brown</b>                 | ear      | seems to be leaping and barking in ineffectual remonstrance with   |
| 875  | tenderly coaxing the crumbled leaf of his cigar with <b>cautious</b>       | fingers. | ¶ "My friend at the corner of Chancery Lane has not                |

Figure 29: Concordance with ADJ on offset -1, sorted alphabetically on -1.

## 9. Groups based on sequences: Partition by N-grams

### Background:

N-grams are sequences of a fixed length. By setting  $n$  to a numeric value, you can change that length to one that suits your research question. A common choice are bigrams ( $n = 2$ ) and trigrams ( $n = 3$ ), as these often point to useful patterns across different texts.

In FlexiConc, n-grams are offered as a **partitioning** algorithm, which creates groups with frequency counts, similar to [section 5](#).

While our example below shows traditional n-grams, you can also use this function to create **skip-grams** (e.g. you could choose positions -1 and 1 to see repeated patterns of preceding and following tokens irrespective of the node term).

As is the case with *Select* and *Select by Token-Level Attribute* ([sections 4](#) and [8](#)), you can also use annotations like POS ([section 3.1](#)) to create n-grams. To do this, change the *word* attribute to the token-level annotation layer that you want to use (*pos\_* for POS tags).

In this example, we use word-level 2-grams (also known as *bigrams*) to get a more comprehensive overview of the adjectives used on L1 and the body part noun used in the node.

### ! Note:

In the CLiC interface outside of FlexiConc mode, n-grams are called *clusters*. We decided to use different terminology to avoid overlap with our clustering algorithm ([step 5](#)).

### Instructions:

1. Select the algorithm **Partition by N-grams** (Figure 30).
2. Add **-1** and **0** to the list of positions to create bigrams for the word on -1 and the node (Figure 31).

Find and use Partition by N-grams like this:

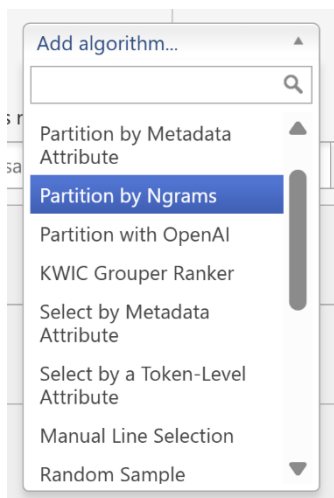


Figure 28: *Partition by N-grams*

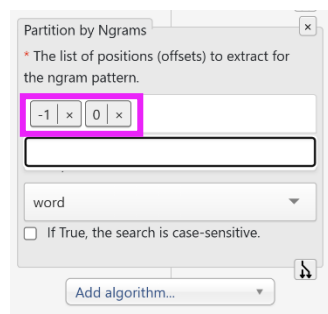


Figure 29: *Partition by N-grams*: Parameter settings

This is what your result should look like:

| Partition                                                           | cheeks cheek neck fingers ear | ('pale', 'cheeks')                                           | 9 lines |
|---------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------|---------|
| she was, the girl's Roman blood rushed over her pale cheeks         |                               | as she felt the insult. She turned towards Hermanric, looked | Antoni  |
| lays. The cold fingers that trembled round mine--the pale cheeks    |                               | with a bright red spot burning in the midst of               | wwhite  |
| inted palace to their appreciative taste--Sue's usually pale cheeks |                               | reflecting the pink of the tinted roses at which she         | Jude    |
| way. I in love--and with him too! Her pale cheeks                   |                               | suddenly became one flame of fire; and she covered her       | NorthS  |
| now was sunken pallor. But the children kissed the pale cheeks      |                               | and never found them deficient. That love was now the        | Deronda |
| 'ou saw Miss Harleth?' " " _She_ told you that?" " The pale cheeks  |                               | looked even paler, perhaps from the fierce brightness in the | Deronda |
| the fresh breeze had braced her up, and her pale cheeks             |                               | were really more rosy. Her mother rejoiced when she saw      | dracula |
| through the window and gave more colour to the pale cheeks.         |                               | When he saw her, Arthur was simply choking with emotion      | dracula |
| tears, and caused them to fall quietly, down her pale cheeks.       |                               | " It was the longest day she had ever known in               | mary    |
| Partition                                                           | cheeks cheek neck fingers ear | ('little', 'fingers')                                        | 6 lines |
| Partition                                                           | cheeks cheek neck fingers ear | ('pink', 'cheeks')                                           | 5 lines |
| Partition                                                           | cheeks cheek neck fingers ear | ('pale', 'cheek')                                            | 4 lines |
| Partition                                                           | cheeks cheek neck fingers ear | ('small', 'fingers')                                         | 4 lines |

Figure 30: Concordance with *ADJ* on position -1, partitioned by n-grams

## 10. Choosing what matters: Manual Line Selection

### Background:

With *Manual Line Selection*, you can create a subset based on lines that you choose. This is especially useful for partitions, as Manual Line Selection lets you pick out one partition to analyse it in more detail.

In the example below, the manual line selection is based on the bigrams created using *Partition by Ngrams* (see [section 9](#)). We will use manual selection to group lines together that share some token positions, but are not entirely the same. We will focus on partitions that indicate cheek complexion. For each partition, the token on -1 is an adjective like *pale* and the node is *cheek* or *cheeks*.

Manual line selection has two modes:

1. you can select **line IDs** to select individual lines, or
  2. **group identifiers** (the label or number of the partitions to choose).
- In our example, we will use this second option to select the entire partition.

### Instructions:

1. Select the algorithm **Manual line selection** (Figure 31).
2. **Click on the group identifier window** (Figure 32).
3. Choose the partitions 'pale cheeks,' 'pink cheeks,' and 'pale cheek' (Figure 33).
4. Confirm by clicking **OK** in the bottom left (Figure 33).

## Find and use Manual Line Selection like this:

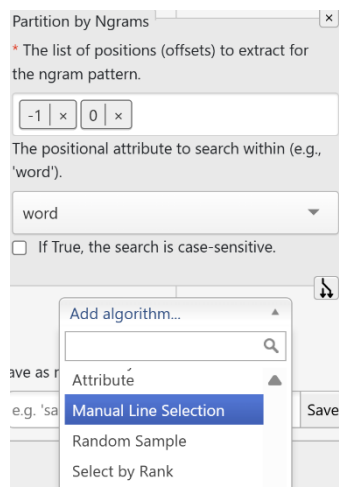


Figure 31: Manual Line Selection

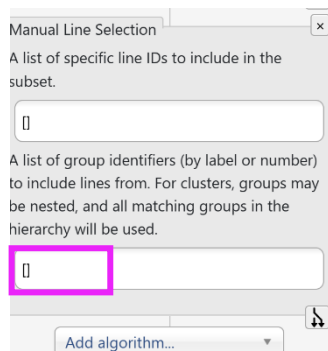


Figure 32: Manual Line Selection: Parameter settings

## Select the partitions like this:

| ID                                    | Left                                                                    | Node                          | Right                                                        | Book    | In bk.  |
|---------------------------------------|-------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------|---------|---------|
| ▼ <input checked="" type="checkbox"/> | Partition                                                               | cheeks cheek neck fingers ear | ('pale', 'cheeks')                                           |         | 9 lines |
| 55                                    | she was, the girl's Roman blood rushed over her pale cheeks             |                               | as she felt the insult. She turned towards Hermanric, looked | Antoni  |         |
| 60                                    | days. The cold fingers that trembled round mine--the pale cheeks        |                               | with a bright red spot burning in the midst of               | wwhite  |         |
| 114                                   | enchanted palace to their appreciative taste--Sue's usually pale cheeks |                               | reflecting the pink of the tinted roses at which she         | Jude    |         |
| 143                                   | way. I in love--and with him too! Her pale cheeks                       |                               | suddenly became one flame of fire; and she covered her       | NorthS  |         |
| 160                                   | now was sunken pallor. But the children kissed the pale cheeks          |                               | and never found them deficient. That love was now the        | Deronda |         |
| 161                                   | you saw Miss Harleth?" "I _She_ told you that?" "I The pale cheeks      |                               | looked even paler, perhaps from the fierce brightness in the | Deronda |         |
| 185                                   | the fresh breeze had braced her up, and her pale cheeks                 |                               | were really more rosy. Her mother rejoiced when she saw      | dracula |         |
| 190                                   | through the window and gave more colour to the pale cheeks.             |                               | When he saw her, Arthur was simply choking with emotion      | dracula |         |
| 270                                   | tears, and caused them to fall quietly, down her pale cheeks.           |                               | "It was the longest day she had ever known in                | mary    |         |
| ▶ <input type="checkbox"/>            | Partition                                                               | cheeks cheek neck fingers ear | ('little', 'fingers')                                        |         | 6 lines |
| ▶ <input checked="" type="checkbox"/> | Partition                                                               | cheeks cheek neck fingers ear | ('pink', 'cheeks')                                           |         | 5 lines |
| ▼ <input checked="" type="checkbox"/> | Partition                                                               | cheeks cheek neck fingers ear | ('pale', 'cheek')                                            |         | 4 lines |

Figure 33: Manually selecting partitions

| ID  | Left                                                                | Node    | Right                                                        | Book    | In bk.  |
|-----|---------------------------------------------------------------------|---------|--------------------------------------------------------------|---------|---------|
| ▼   | Partition cheeks cheek neck fingers ear ('pale', 'cheeks')          |         |                                                              |         | 9 lines |
| 35  | she was, the girl's Roman blood rushed over her pale                | cheeks  | as she felt the insult. She turned towards Hermanric, looked | Antoni  |         |
| 60  | days. The cold fingers that trembled round mine--the pale           | cheeks  | with a bright red spot burning in the midst of               | wwhite  |         |
| 114 | an enchanted palace to their appreciative taste--Sue's usually pale | cheeks  | reflecting the pink of the tinted roses at which she         | Jude    |         |
| 143 | way. I in love--and with him too! Her pale                          | cheeks  | suddenly became one flame of fire; and she covered her       | NorthS  |         |
| 160 | now was sunken pallor. But the children kissed the pale             | cheeks  | and never found them deficient. That love was now the        | Deronda |         |
| 161 | You saw Miss Harleth?" "I " _She_ told you that?" "I The pale       | cheeks  | looked even paler, perhaps from the fierce brightness in the | Deronda |         |
| 186 | the fresh breeze had braced her up, and her pale                    | cheeks  | were really more rosy. Her mother rejoiced when she saw      | dracula |         |
| 190 | through the window and gave more colour to the pale                 | cheeks. | When he saw her, Arthur was simply choking with emotion      | dracula |         |
| 270 | tears, and caused them to fall quietly, down her pale               | cheeks. | "It was the longest day she had ever known in                | mary    |         |
| ►   | Partition cheeks cheek neck fingers ear ('pink', 'cheeks')          |         |                                                              |         | 5 lines |
| ►   | Partition cheeks cheek neck fingers ear ('pale', 'cheek')           |         |                                                              |         | 4 lines |

Figure 34: Manually selected partitions

**This is what your results should look like:**

## 11. A bit of everything: Random Sample

**Background:** Random samples are useful when

- the concordance is too large to analyse fully, or
- if you want to generate hypotheses before investigating them in the overall corpus

In [section 2](#), we applied Random Sort, which retained all lines in the concordance and showed them in a semi-random order.

In contrast, Random Sample performs random sorting in the background and then creates a subset with the number of lines specified by the user.

### Instructions:

1. Create a new branch in the tree using the + button in the branch overview (Figure 35).
2. Select the **Random Sample** algorithm. (Figure 36).
3. Set the **number of lines to sample** to **500** (Figure 37).
4. Keep the **random seed**, which ensures a reproducible sample (Figure 37).

### ! Notes:

1. You should always set a **seed** to make your random sampling reproducible.
2. Creating a **branch** via the + sign results in a *top-level branch*. None of the algorithms used so far are carried over – branch 3 is an entirely blank slate, while branches 1 & 2 are left intact.

This is how to find and use *Random Sample*:

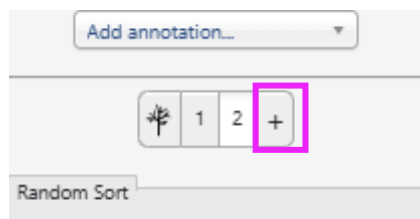


Figure 35: Creating a new top-level branch

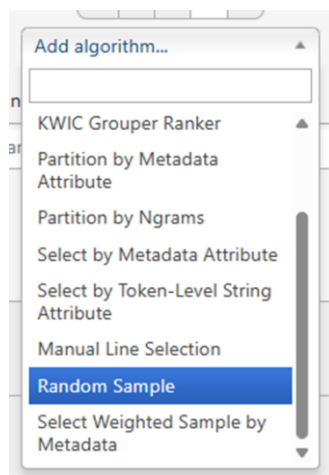


Figure 36: *Random Sample*

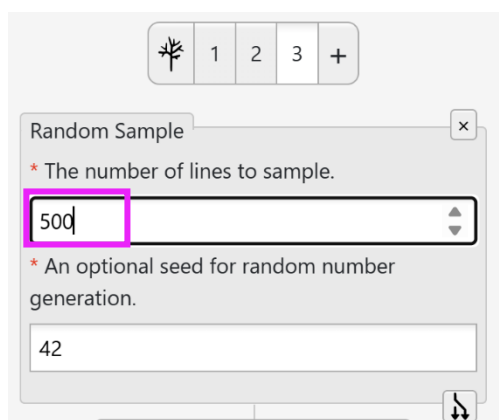


Figure 37: *Random Sample*: Parameter settings

This is what your result should look like:

Showing 1 to 50 of 500 entries,

54b4590

| ID | Left                                                           | Node    | Right                                                           | Book    | In bk.      |
|----|----------------------------------------------------------------|---------|-----------------------------------------------------------------|---------|-------------|
| 1  | at half-past nine o'clock, singing a little Scotch melody, her | cheeks  | tinged with as delicate a pink as the pale hue                  | LadyAud | <div></div> |
| 3  | of that sunless, flaxen hue which generally accompanies pink   | cheeks  | and white eyelashes. She had been a rustic beauty, perhaps      | LadyAud | <div></div> |
| 4  | clinging about him, kissed the dirty tears from his faded      | cheeks. | "Don't be sorry for me, gran'pa," he said; "I am                | LadyAud | <div></div> |
| 6  | better nor half the folks as are grown up." "Maggie's          | cheeks  | began to flush with triumphant excitement. She thought Mr. mill |         | <div></div> |
| 7  | lavishing so many warm kisses on it that the waxen             | cheeks  | had a wasted, unhealthy appearance. "Did you ever hear th       | mill    | <div></div> |
| 13 | and still Tom laughed and clapped his hands, and Maggie's      | cheeks  | began to pale, and her lips to tremble a little                 | mill    | <div></div> |
| 14 | turn towards the same point as her own, and Maggie's           | cheeks  | and ears began to burn, while uncle Glegg, a kind-looking       | mill    | <div></div> |
| 16 | Columbus?" "Maggie's eyes had begun to sparkle and her         | cheeks  | to flush,--she was really beginning to instruct the gypsies     | mill    | <div></div> |
| 19 | ne ambitious. The exercise brought the warm blood into her     | cheeks, | and made her inclined to take her lesson merrily. "I            | mill    | <div></div> |
| 23 | gone before, a glowing gladness at heart. Her eyes and         | cheeks  | were still brightened with her childlike enthusiasm in the da   | mill    | <div></div> |

Figure 38: Concordance of a *Random Sample* with 500 lines

This handout builds on work funded by the "Reading Concordances in the 21st Century" research project supported by the Arts and Humanities Research Council (AHRC) (grant references: AH/X002047/1 & AH/X002047/2) and the Deutsche Forschungsgemeinschaft (DFG) (grant reference: 508235423).  
Using FlexiConc via CLIC: a case study on literary fiction © 2026 by Nathan Dykes, Stephanie Evert, Michaela Mahlberg, Alexander Piperski is licensed under [CC BY-NC 4.0](https://creativecommons.org/licenses/by-nc/4.0/)



## 12. Information about the text: Select by Metadata Attribute

### Background:

**Metadata** is information *about* the text rather than what is *in* the text (the concordance lines).

Which metadata attributes are available and what they are called may vary between corpora.

In the 19C corpus, available metadata includes:

- the book title (coded as **text\_id**)
- chapter number, and
- further structural information.

Like Select by Token-Level String Attribute ([section 4](#)), Select by Metadata Attribute creates a subset of the original concordance. As we are dealing with a corpus of novels, Select by Metadata Attribute lets us focus on concordance lines in specific books.

### Instructions:

1. Find *Select by Metadata Attribute* in the *add algorithm* menu (Figure 39).
2. Set the **metadata attribute** to **text\_id** (Figure 40).
3. Set the **value to compare against** to **Deronda** (Figure 41).

### Find and use Select by Metadata Attribute like this:

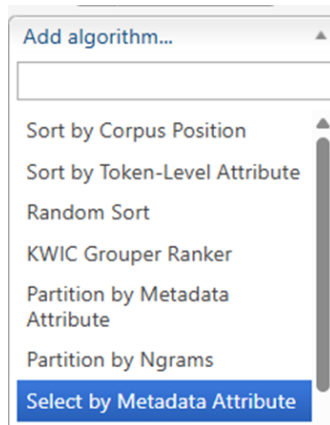


Figure 39: *Select by Metadata Attribute*

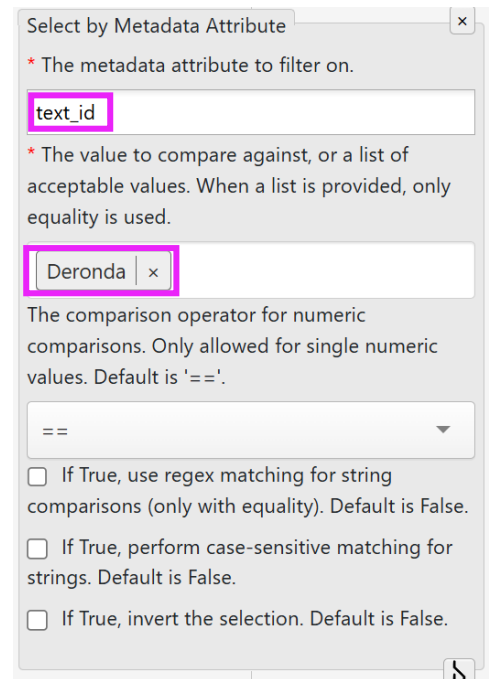


Figure 40: *Select by Metadata Attribute: Parameter settings*

### This is what your result should look like:

Showing 1 to 45 of 45 entries,

| ID  | Left                                                            | Node    | Right                                                         | Book    | In bk. |
|-----|-----------------------------------------------------------------|---------|---------------------------------------------------------------|---------|--------|
| 148 | with conflict. It did not bring the blood to her                | cheeks, | but it sent it away from her lips. She controlled             | Deronda |        |
| 150 | longed to somebody else. Her eyes had become brighter, her      | cheeks  | slightly flushed, and her tongue ready for any mischievous re | Deronda |        |
| 154 | followed hers. The handsome little fellow was puffing out his   | cheeks  | in trying to blow a tiny trumpet which remained dumb          | Deronda |        |
| 161 | You saw Miss Harleth?" "I "She told you that?" "I The pale      | cheeks  | looked even paler, perhaps from the fierce brightness in the  | Deronda |        |
| 163 | of his pockets; or drawing down the skin of his                 | cheeks  | to make his eyes look awful, and rolling his head             | Deronda |        |
| 164 | c-stool without speaking, and the tears were rolling down her   | cheeks  | as she looked at her friends. "Now, now, Mab!" said           | Deronda |        |
| 167 | her son's shoulders at once, and kissed him on both             | cheeks, | then seated herself among her cushions with an air of         | Deronda |        |
| 392 | ch by giving her sisters lessons." Here Mrs. Davilow's delicate | cheek   | showed a rapid blush. "If it were not for that                | Deronda |        |
| 393 | figure, her long white throat, and the curves of her            | cheek   | and chin were always set off to perfection by the             | Deronda |        |
| 395 | after the long sorrow and watching which had paled her          | cheek   | and made blue semicircles under her eyes. It was Mab          | Deronda |        |
| 403 | knew you would be." Then she kissed him on each                 | cheek,  | and he returned the kisses. But it was something like         | Deronda |        |
| 407 | from her foretaste of joy; some couple, bending cheek by        | cheek,  | over a bit of work done by the one and                        | Deronda |        |
| 408 | She was seated gazing fixedly on the sea, resting her           | cheek   | on her hand, looking less shattered than when he had          | Deronda |        |
| 668 | in an infallible system which placed his foot on the            | neck    | of chance--immediately prepared a new pile. So did a          | Deronda |        |

Figure 41: *Select by Metadata Attribute: Concordance lines from the novel Deronda*

This handout builds on work funded by the "Reading Concordances in the 21st Century" research project supported by the Arts and Humanities Research Council (AHRC) (grant references: AH/X002047/1 & AH/X002047/2) and the Deutsche Forschungsgemeinschaft (DFG) (grant reference: 508235423). Using FlexiConc via CLIC: a case study on literary fiction © 2026 by Nathan Dykes, Stephanie Evert, Michaela Mahlberg, Alexander Piperski is licensed under [CC BY-NC 4.0](https://creativecommons.org/licenses/by-nc/4.0/)



## 13. Grouping by books or chapters: partition by metadata

### Background:

In this step, we use **metadata** – information *about* the text rather than what is *in* the text (the concordance lines) – to group concordance lines.

Partition by Metadata creates partitions of the type that were also used in [step 5](#) (Flat clustering) and [step 9](#) (Partition by Ngrams). By partitioning based on metadata, we can see how patterns are distributed across all novels in the text, or across all chapters in a specific book. In this example, we apply chapter partitions to our *Deronda* subset from [section 12](#) to see whether body part nouns are more common in specific chapters of the novel.

### Instructions:

1. Find *Partition by Metadata* in the *add algorithm* menu (Figure 42)
2. Set the metadata attribute to **chapter** (Figure 43).
3. Keep the checkbox ticked to see partitions sorted from largest to smallest (Figure 43).

Find and use Partition by Metadata Attribute like this:

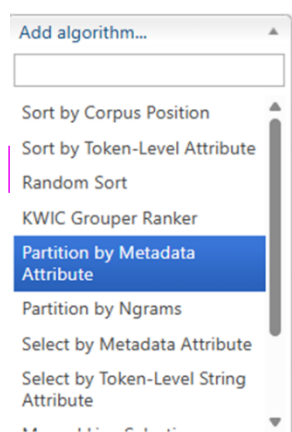


Figure 42: *Partition by Metadata Attribute*

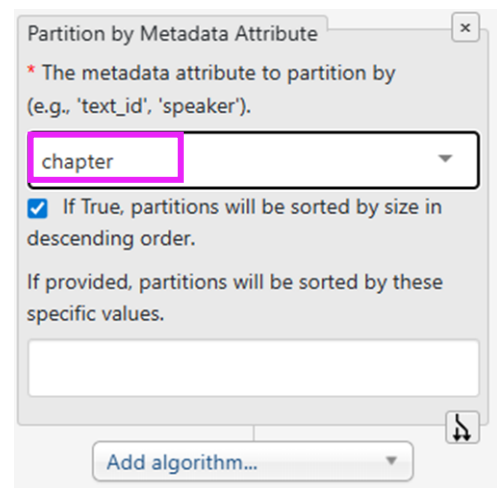


Figure 43: *Partition by Metadata Attribute: Parameter settings*

### This is what your result should look like:

| Partition                                               | cheeks cheek neck fingers ear | 48                                                                    | 12 lines |
|---------------------------------------------------------|-------------------------------|-----------------------------------------------------------------------|----------|
| boy." ¶ Gwendolen's lips were almost as pale as her     | cheeks;                       | her passion had no weapons--words were no better than                 | Deronda  |
| and from the yoke she had brought on her own            | neck.                         | No! she foresaw him always living, and her own life                   | Deronda  |
| that she should die with his throttling fingers on her  | neck                          | avenging that thought. Fantasies moved within her like ghosts, making | Deronda  |
| black dress her head on its white pillar of a           | neck                          | showed to advantage. Some consciousness of this made her turn         | Deronda  |
| her crown of hair so as completely to conceal her       | neck,                         | and leave only her face looking out from the black                    | Deronda  |
| z, her husband was arriving with a cigar between his    | fingers.                      | He threw it away and handed her out, accompanying her                 | Deronda  |
| nto the terror that she should die with his throttling  | fingers                       | on her neck avenging that thought. Fantasies moved within her         | Deronda  |
| f with the jewels that glittered on her tightly-clasped | fingers                       | pressed against her heart. ¶ The feeling Deronda endured in these     | Deronda  |
| hand, and she let him press her poor little chill       | fingers;                      | but she said no good-bye. ¶ When he had left the                      | Deronda  |
| of the right shape; nor one the lobe of whose           | ear                           | was at all too large and red; nor one who                             | Deronda  |
| le sneer in his low voice which in poor Gwendolen's     | ear                           | was diabolical. ¶ For the first time since their marriage she         | Deronda  |
| ad no intention to the contrary, but to Gwendolen's     | ear                           | his words had as much insolence in them as his                        | Deronda  |
| Partition                                               | cheeks cheek neck fingers ear | 30                                                                    | 9 lines  |
| Partition                                               | cheeks cheek neck fingers ear | 23                                                                    | 7 lines  |
| Partition                                               | cheeks cheek neck fingers ear | 29                                                                    | 7 lines  |
| Partition                                               | cheeks cheek neck fingers ear | 39                                                                    | 6 lines  |

Figure 44: *Partition by Metadata Attribute*: Concordance lines from the novel *Deronda* across chapters.

#### ! Note:

As you can see in Figure 46, *Deronda* has many chapters, making it difficult to spot any tendencies right away.

💡 If you see any partitions or lines that interest you, use **Manual Line Selection** ([section 10](#)) to create different subsets from a specific range of chapters (e.g. the first and second half of the novel).

## 14. Maintaining balance: Select Weighted Sample by Metadata

### Background:

The *Select Weighted Sample by Metadata* creates a subset of the concordance that contains the same proportion of lines per book or chapter as in the overall concordance.

To create this sample, the user specifies:

- the **number of lines** to draw, similar to Random Sample ([section 11](#))
- a **metadata attribute** to weight by, such as *text\_id* (book title).

Compared to Random Sample, which treats all lines as equally likely, the weighted sample will create a sample that is more similar to the overall concordance in terms of the specified attribute.

**! Note:** As with any semi-random operation, you should use a **seed** to ensure your selection is reproducible.

### Instructions:

1. Create a new empty **branch** using the plus sign (+) next to the tree symbol (Figure 45).
2. Find *Select Weighted Sample by Metadata* in the *add algorithm* menu (Figure 46).
3. Create a sample based on the attribute **text\_id** (the book title, Figure 47).
4. Set the **total number of lines** to **500** (Figure 47).
5. Keep the **random seed** intact (Figure 47).

Find and use **Select Weighted Sample by Metadata** like this:

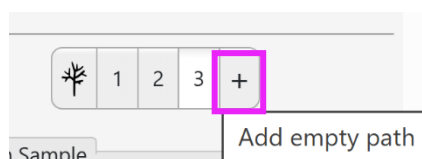


Figure 45: Create an empty branch

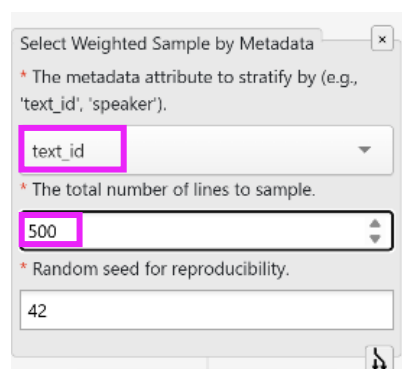


Figure 47: *Select Weighted Sample by Metadata*: Parameter Settings

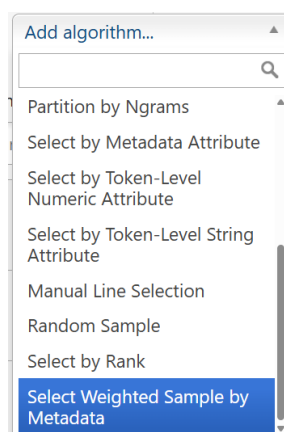


Figure 46: *Select Weighted Sample by Metadata*

This is what your result should look like:

| Left                                                              | Node    | Right                                                            |
|-------------------------------------------------------------------|---------|------------------------------------------------------------------|
| were of that sunless, flaxen hue which generally accompanies pink | cheeks  | and white eyelashes. She had been a rustic beauty, perhaps       |
| better nor half the folks as are grown up." ¶ Maggie's            | cheeks  | began to flush with triumphant excitement. She thought Mr. Riley |
| lavishing so many warm kisses on it that the waxen                | cheeks  | had a wasted, unhealthy appearance. ¶ "Did you ever hear the     |
| turn towards the same point as her own, and Maggie's              | cheeks  | and ears began to burn, while uncle Glegg, a kind-looking        |
| Maggie, my dear; don't make yourself so ugly!" and Tom's          | cheeks  | were looking particularly brilliant as a relief to his best      |
| about Columbus?" ¶ Maggie's eyes had begun to sparkle and her     | cheeks  | to flush,—she was really beginning to instruct the gypsies       |
| with a pair of amiable blue-gray eyes and round pink              | cheeks  | that refused to look formidable, let him frown as he             |
| chair, with the big tears ready to roll down her                  | cheeks, | quite blinded to the presence of Bob, who was looking            |
| was the instinctive vent of strong excitement. Her eyes and       | cheeks  | had an almost feverish brilliancy; her head was thrown backward  |
| must have come again by the river. Maggie felt her                | cheeks  | glowing and her heart beating; it was natural she should         |
| he had lashed himself into fury. His eyes glared, his             | cheeks  | flushed, his voice rose. Could he then have seen the             |

Figure 48: Concordance for *Select Weighted Sample by Metadata*

## 15. Ranking by pattern counts: KWICGrouper Ranker

### Background:

**KWICGrouper Ranker** is a variant of KWICGrouper (O'Donnell 2008). KWICGrouper Ranker reorders the concordance lines based on how many matches of specified search terms they contain.

Compared to the KWICGrouper implementation in the traditional CLiC interface, KWICGrouper Ranker features some extra functionality, such as switching between type (every form is only counted once) and token counts (every match is sorted, regardless of repetitions).

In the example, we will rank the concordance lines based on the number of *-ing* forms.

**! Note:** If you have annotated your concordance with POS tags ([section 3.1](#)), you can also KWICGroup on the PoS level, for instance, to rank by the number of adjectives (ADJ).

In that case, you should disable the checkbox for type-counting to make sure that more than one ADJ tag per line can be counted.

### Instructions:

1. If you are following the overall example, add this algorithm to the Weighted Random Sample created in **branch 4** (Figure 45).
2. Find *KWICGrouper Ranker* in the *add algorithm* menu (Figure 49).
3. Set the **matching strategy** to *cqp* (Figure 50)
4. Enter **.+ing** as a search term to find words that end in *ing* (Figure 50)

Find and use KWICGrouperRanker like this:

Select Weighted Sample by Metadata

\* The metadata attribute to stratify by (e.g., 'text\_id', 'speaker').

text\_id

\* The total number of lines to sample.

500

An optional seed for generating the pseudo-random order.

42

Save

Figure 49: KWIC Grouper Ranker

KWIC Grouper Ranker

\* A list of terms to search for within the tokens.

.+ing

word

Matching strategy for search\_terms

cqp

☐ If True, the search is case-sensitive.

☐ If True, include node-level tokens in the search.

The lower bound of the window (offset range).

The upper bound of the window (offset range).

☒ If True, count unique types within each line; otherwise, count all matches.

Figure 50: KWIC Grouper Ranker: Parameter Settings

**! Note:** The search term **.+ing** that we ranked on is a **regular expression**. Regex allow you to specify patterns -ing forms regardless of which words they are. By using **cqp** mode over **regex** mode, token boundaries are automatically assumed - **.+ing** in traditional regex mode would also retrieve *thingamy* unless you specify an explicit word boundary **.+ing \$**

**.+** means that the word can start with any character and be of arbitrary length; but there must be at least one character before *ing*.

**\$** means that the word has to end after *-ing*, so we do not get matches like *thingamy*. We will, however, still get some unintended results (e.g. *something*).

### This is what your result should look like:

|                                                                                                                                                    |                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| not here?' she demanded, <b>pulling</b> off her gloves, and <b>displaying</b> fingers                                                              | wonderfully whitened with <b>doing nothing</b> and <b>staying</b> indoors. ¶ 'Heathcliff, you |
| work, and went on <b>making</b> false stitches and <b>pricking</b> her fingers                                                                     | with much perseverance, not <b>looking</b> up or <b>taking</b> notice of                      |
| who was <b>speaking</b> with animation--now <b>stretching</b> out his long fingers                                                                 | horizontally, now <b>pointing</b> downward with his fore-finger, now <b>folding</b> his       |
| instant on the <b>shadowing</b> eye-lashes before <b>rolling</b> slowly down her cheeks,                                                           | and <b>dropping</b> , unheeded, on her dress. ¶ They were to stop                             |
| <b>rushing</b> into the room, threw his arms round the Prince's neck. ¶ "My <b>darling!</b> " said the father, <b>forgetting</b> at this moment of |                                                                                               |
| much for the sables," she said, <b>holding</b> out her little fingers,                                                                             | all <b>glittering</b> and <b>twinkling</b> with the diamonds she wore upon                    |
| He looked vexed and puzzled, and sat <b>drumming</b> with his fingers                                                                              | on the dining-table without <b>uttering</b> a word. ¶ "I'm afraid <b>something</b>            |
| her back towards them, was <b>counting</b> her change with <b>trembling</b> fingers,                                                               | not <b>daring</b> to turn round until she heard the answer                                    |
| <b>spilling</b> their milk over the table and themselves, <b>plunging</b> their fingers                                                            | into their own or each other's mugs, or <b>quarrelling</b> over                               |
| jewels from her arms, and the rings from her <b>shaking</b> fingers,                                                                               | and held them all in a heap, <b>quivering</b> and <b>looking</b>                              |
| Mrs. Tulliver, <b>beckoning</b> Maggie to her, and <b>whispering</b> in her ear,                                                                   | as soon as this point of Lucy's <b>staying</b> was settled                                    |
| must have come again by the river. Maggie felt her cheeks                                                                                          | <b>glowing</b> and her heart <b>beating</b> ; it was natural she should                       |
| like to hear it." ¶ Midwinter suddenly looked up again, his cheeks                                                                                 | <b>turning</b> ashy pale, and his <b>glittering</b> black eyes fixed full                     |
| spoke these last words the tears were <b>running</b> down his cheeks,                                                                              | and he was <b>weeping</b> openly in presence of his companion                                 |

Figure 51: Concordance for *KWICGrouperRanker*

**! Note:** KWICGrouperRanker creates a new column **Ranking**, which shows the number of matches for each concordance line: the first line contains 5 -ing words, followed by two lines with four matches (Figure 52).

|  | Book    | In bk.                                                                              | Ranking:<br>KWIC<br>Grouper<br>Ranker |
|--|---------|-------------------------------------------------------------------------------------|---------------------------------------|
|  | wh      |  | 5                                     |
|  | mill    |  | 4                                     |
|  | Deronda |  | 4                                     |
|  | NorthS  |  | 3                                     |
|  | VivianG |  | 3                                     |
|  | LadyAud |  | 3                                     |
|  | arma    |  | 3                                     |
|  | NorthS  |  | 3                                     |
|  | AgnesG  |  | 3                                     |
|  | vanity  |  | 3                                     |

Figure 52: Rank score for *KWICGrouperRanker*

## 16. Wrapping up: saving and loading your work

### Background:

Besides being useful for analysis, the tree that FlexiConc creates ([section 6](#) and [section 7](#)) serves reproducibility and long-term transparency.

You can save your tree to a JSON file and load the same file into FlexiConc at a later stage. This allows you to:

- pause your analysis and resume it later.
- share your work with a colleague.
- append your analysis to a journal article or term paper.
- trace other analysts' steps to understand their procedure.

### Instructions:

1. Go to the **analysis tree** view by clicking on the tree symbol (Figure 53).
2. **Click on save to file.** While in tree view mode, you can find this button right below the tree symbol (Figure 54).
3. The tree is automatically downloaded. Find it in your downloads folder and name it **rc21\_flexiclic\_demo.json**
4. Click on load from file and select the tree file for upload. FlexiConc should start reloading your analysis.

This is how to find the tree and the save/load functions



Figure 53: Accessing the analysis tree.

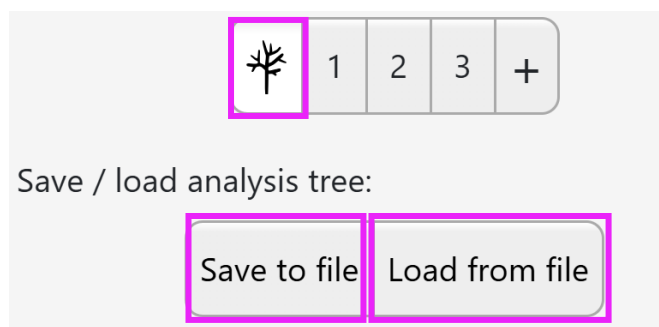


Figure 54: Save and load buttons.



! By default, the file is downloaded as clic-analysis-tree. json. **You should always rename your file and move it to the folder where you keep your project!**

1. Name your tree **rc21\_flexiclic\_demo.json**
2. Save it in a location where you will be able to find it again!

## References

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